

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

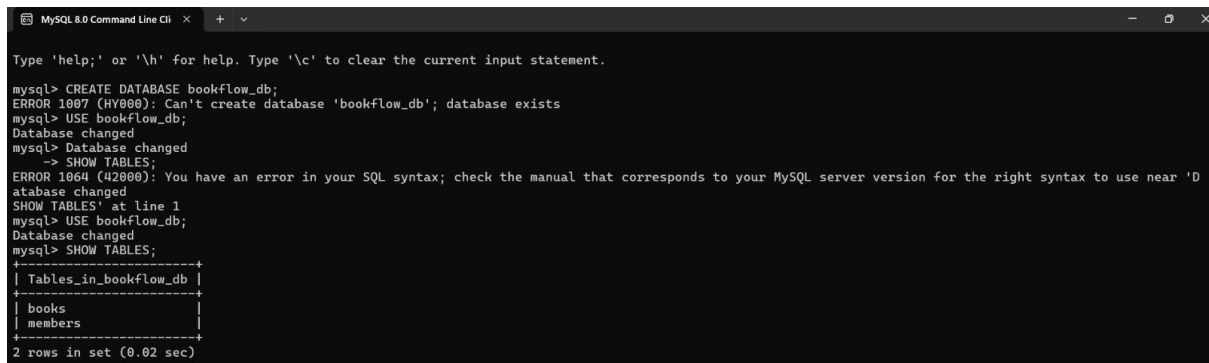
Week-1

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Step 1 :Database Setup Query 1: Create the database

CREATE DATABASE IF NOT EXISTS bookflow_db;

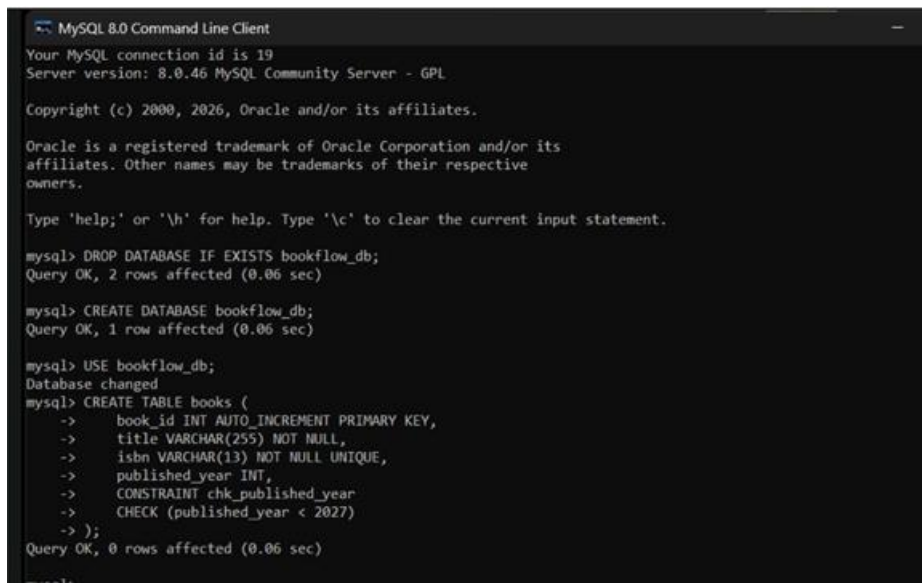


```
MySQL 8.0 Command Line Client
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bookflow_db;
ERROR 1007 (HY000): Can't create database 'bookflow_db'; database exists
mysql> USE bookflow_db;
Database changed
mysql> Database changed
-> SHOW TABLES;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'D
atabase changed
SHOW TABLES' at line 1
mysql> USE bookflow_db;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db |
+-----+
| books                   |
| members                 |
+-----+
2 rows in set (0.02 sec)
```

Query 2: Select the database

USE bookflow_db;



```
MySQL 8.0 Command Line Client
Your MySQL connection id is 19
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> DROP DATABASE IF EXISTS bookflow_db;
Query OK, 2 rows affected (0.06 sec)

mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.06 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE books (
  ->   book_id INT AUTO_INCREMENT PRIMARY KEY,
  ->   title VARCHAR(255) NOT NULL,
  ->   isbn VARCHAR(13) NOT NULL UNIQUE,
  ->   published_year INT,
  ->   CONSTRAINT chk_published_year
  ->   CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
```

3. Step 2 Table

Creation (DDL) with Constraints

Query 3: Create the books table

```
mysql> INSERT INTO books (title, isbn, published_year) VALUES
-> ('The Alchemist', '9780061122415', 1988),
-> ('Clean Code', '9780132350884', 2008),
-> ('Atomic Habits', '9780735211292', 2018);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM books;
+-----+-----+-----+-----+
| book_id | title      | isbn      | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist | 9780061122415 | 1988          |
| 2       | Clean Code    | 9780132350884 | 2008          |
| 3       | Atomic Habits | 9780735211292 | 2018          |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Query 4: Create the members table

```
mysql> INSERT INTO members (full_name, email) VALUES
-> ('Anil Kumar', 'anil.kumar@example.com'),
-> ('Priya Sharma', 'priya.sharma@example.com'),
-> ('Ravi Verma', 'ravi.verma@example.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM members;
+-----+-----+-----+
| member_id | full_name | email                      |
+-----+-----+-----+
| 1         | Anil Kumar | anil.kumar@example.com    |
| 2         | Priya Sharma | priya.sharma@example.com  |
| 3         | Ravi Verma  | ravi.verma@example.com    |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

Query 5 (verification): Inspect the table structures

```
mysql> DESCRIBE books;
```

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(255)	NO		NULL	
isbn	varchar(13)	NO	UNI	NULL	
published_year	int	YES		NULL	

```
4 rows in set (0.00 sec)
```

```
mysql> DESCRIBE members;
```

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

```
3 rows in set (0.01 sec)
```

```
mysql> |
```

Query 6: Insert 3 books

```
INSERT INTO books (title, isbn, published_year) VALUES
('The Alchemist', '9780061122415', 1988),
('Clean Code', '9780132350884', 2008),
('Atomic Habits', '9780735211292', 2018);
```

Query 7: Insert 3 members

```
INSERT INTO members (full_name, email) VALUES
('Anil Kumar', 'anil.kumar@example.com'),
('Priya Sharma', 'priya.sharma@example.com'),
('Ravi Verma', 'ravi.verma@example.com');
```

Test 1: Duplicate ISBN (UNIQUE violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9780061122415', 2000);
```

Output:

Explanation: The ISBN 9780061122415 already belongs to 'The Alchemist'. The UNIQUE constraint on isbn rejects the row, so two physical records can never claim the same ISBN.

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Output:

Explanation: The NOT NULL constraint fires because a book must have a name. The database refuses to store a nameless book, exactly as the requirement demands.

Test 3: Future publication year (CHECK violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Time Traveler', '8888888888888', 2030);
```

Output:

Explanation: 2030 fails the condition `published_year < 2027`, so the named CHECK constraint `chk_published_year` blocks the insert. Garbage dates cannot enter the table.

Test 4: Duplicate email (UNIQUE violation)

```
INSERT INTO members (full_name, email)
VALUES ('Anil Clone', 'anil.kumar@example.com');
```

Explanation: `anil.kumar@example.com` is already registered to member_id 1, so the UNIQUE constraint on email rejects the second signup. One email = one account

```
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES ('Fake Copy', '9780061122415', 2000);
ERROR 1062 (23000): Duplicate entry '9780061122415' for key 'books.isbn'
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES (NULL, '9999999999999', 2010);
ERROR 1048 (23000): Column 'title' cannot be null
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES ('Time Traveler', '8888888888888', 2030);
ERROR 3819 (HY000): Check constraint 'chk_published_year' is violated.
mysql> INSERT INTO members (full_name, email)
-> VALUES ('Anil Clone', 'anil.kumar@example.com');
ERROR 1062 (23000): Duplicate entry 'anil.kumar@example.com' for key 'members.email'
```